

Two-time Nobel laureate Frederick Sanger was an experimental biochemist who sequenced the components of the three fundamental polymers of life: proteins, RNA, and DNA. His technique for sequencing DNA molecules revolutionized molecular biology and heralded a new era in medicine, gene therapy, and genetic manipulation.

MILESTONES

DOCTORAL RESEARCH

Awarded PhD in 1943 for studies into amino acids, the building blocks of the body's proteins.

NOBEL AWARD

Receives the 1958 Nobel Prize in Chemistry for his work on the chemical sequencing of insulin.

SANGER SEQUENCING

In 1977, publishes his dideoxy method of DNA sequencing, later known as "Sanger Sequencing."

SECOND NOBEL PRIZE

Shares the 1980 Nobel Prize in Chemistry with Walter Gilbert and Paul Berg for DNA sequencing.

RESEARCH CENTER

Opens genome research center the Wellcome Sanger Institute in Cambridge, UK, in 1993.

An ingenious experimentalist.
Sanger developed novel day-to-day laboratory techniques, such as micromethods for manipulating tiny volumes of chemicals.

Frederick Sanger was born in 1918 in Gloucestershire, UK, the son of a doctor who ran a rural general practice. As a child, he was influenced by his brother's interest in natural history, but equally formative were his father's Quaker beliefs, which gave him a strong sense of truth and conscience—philosophies that would later infuse his scientific career. Sanger went to Cambridge University in 1936 to study natural sciences, and it was there that he became interested in biochemistry; he graduated in 1939 with an undergraduate degree.

Scientific path

Sanger then decided to study advanced biochemistry. His parents had both died while he was an undergraduate; using some of the money he had inherited from them, in 1940, he began a self-funded PhD on the metabolism of amino acids, the basic chemical building blocks of proteins. Exempted from military service during World War II due to his Quaker pacifism, he instead conducted applied research on nitrogen uptake in potatoes for the government. He received his PhD in 1943.

Inspired by the collective momentum of experimentation and laboratory work, Sanger later cited the methodology of research as his greatest strength. He joined a group of scientists in Cambridge who were studying proteins and spent the next decade painstakingly attempting to decipher the complete amino acid sequence of insulin. The sequence was completed in 1955. Sanger also revealed that each protein possesses a unique set of amino acids and distinct 3D structure. Insulin

